SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Scientific Rationale and Contract

Scientific contract version:	**v0.1**
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Scientific owner:	Grant Wilson
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Authority boundary. This document specifies an implementation-independent scientific contract. It consolidates the admitted frozen ordinary-map derivation under later owner specializations. It does not claim implementation conformity, representation fidelity, scientific validation, observational performance, or production readiness.

Physical model and contract at a glance

An externally calibrated and conditioned detector sample is a noisy response to a declared Stokes-I sky quantity. An externally supplied projection places each eligible sample on a declared pixel grid. SCI-MAP applies one fixed, positive-coefficient linear averaging operator to the signal and to every declared linear companion. A coadd then applies a second positive-coefficient operator to atomically admitted, compatible observation bundles on the same grid. All scientific statements are conditional on the complete admitted calibration, alignment, eligibility, coefficient, response, geometry, and covariance state.

Facet	Compact authority
Input	Calibrated ordinary-`xs`, Stokes I, `mJy/beam`; explicit identity and eligibility; external coordinates/WCS; finite positive ordinary coefficients; named exposure; response tracer or typed unavailable state.
Output	A complete immutable observation-map bundle and, when requested and admissible, a complete centered-integer common-grid observation coadd bundle.
Equation	On admitted membership, $N_p = \sum_i a_{pi} x_i$ and $Q_p = \sum_i a_{pi}$. The normalized scientific vector contains exactly the rows authorized by the effective support policy, with $\widehat{x}_p = N_p/Q_p$ there; response and covariance use the same explicit row selection. Coadd repeats this restricted structure over complete observations.
Source	Owner-approved Scope Brief; its owner-approved Supersession Cover; exact frozen core Git object `c28f18ed...:SCI-MAP-001_INDEPENDENT_CORE.tex` (SHA-256 `13dd5922...c7324381`); sanitized conventions and ownership.
Status	Contract author draft. Algebraic predictions are preregistered below; none has been executed here. All stronger claim layers remain unassessed.

Contents

1 Scientific boundary and estimand

The ordinary observation map estimates the coefficient-weighted average of the admitted calibrated sample quantity in each declared map pixel. This is a conditional, response-bearing map-domain estimand: its meaning is inherited from SCI-CAL's calibrated input, ALIGN/AST's coordinate relation, PTC's conditioned sample and coefficient identity, and VAL's eligibility state. SCI-MAP transforms those meanings but does not redefine them.

Let $i = (t, d)$ identify a candidate sample occurrence and p a pixel. The effective positive coefficient $a_{pi} = G_{pi}\omega_i$ combines the declared nonnegative projection coefficient with the admitted ordinary analysis coefficient. The fixed contribution set $\mathcal{C}_p$ is resolved before arithmetic. Let Π_{eff} be the recorded effective product support policy and $\mathcal{S}_{\text{out}}(\Pi_{\text{eff}})$ the exact set of map rows for which every support predicate that policy designates applicable passes. This is a row domain, not a numerical mask. The frozen ordinary result is then reused directly:

$$a_{pi} = G_{pi}\,\omega_i, \tag{1}$$

$$\mathcal{C}_p = \{i : \text{all declared contribution predicates pass and } a_{pi} > 0\}, \tag{2}$$

$$\mathcal{S}_{\text{out}}(\Pi_{\text{eff}}) = \{p : \text{authorize}_{\Pi_{\text{eff}}}(p) = 1\}, \tag{3}$$

$$N_p = \sum_{i\in\mathcal{C}_p} a_{pi}x_i, \qquad Q_p = \sum_{i\in\mathcal{C}_p} a_{pi}, \tag{4}$$

$$K_p = \sum_{i\in\mathcal{C}_p} a_{pi}k_i, \tag{5}$$

$$\widehat{x}_p = N_p/Q_p, \qquad \widehat{k}_p = K_p/Q_p, \qquad p \in \mathcal{S}_{\text{out}}, \tag{6}$$

The supported division is not a recipe for filling unsupported pixels. Raw accumulators and representation-specific storage slots may exist outside $\mathcal{S}_{\text{out}}$, but the normalized scientific vector has no row there. A dense zero, sentinel, or coincidentally finite payload is not a semantic substitute and is not measured zero sky. Likewise, Q_p is a normalization fact. Its dimensions or historical name do not make it precision.

The active quantity boundary is `mJy/beam`. A beam-calibrated sample and map require the admitted upstream beam and calibration identities; SCI-MAP does not supply those facts. Accepted tokens for other units do not authorize conversion. V0.1 is Stokes I only and does not authorize measured-R, Q/U, JINC, maximum-likelihood mapmaking, mosaicking, or reprojection.

2 Linear operator, response, and absolute mode

For selected samples $\boldsymbol{y}$, projection matrix G, diagonal coefficient matrix Ω, forward sky-to-sample operator H, and exact effective output-row selector J_{out}, the map operator and its conditional response and covariance are:

$$D_Q = \operatorname{diag}(G\Omega\mathbf{1}), \qquad J_{\text{out}}\boldsymbol{z} = (z_p)_{p\in\mathcal{S}_{\text{out}}}, \qquad D_{Q,\text{out}} = J_{\text{out}} D_Q J_{\text{out}}^{\mathsf{T}}, \tag{7}$$

$$A_{\text{out}} = D_{Q,\text{out}}^{-1} J_{\text{out}} G\Omega, \qquad \widehat{\boldsymbol{x}}_{\text{out}} = A_{\text{out}}\boldsymbol{y}, \tag{8}$$

$$R_{\text{out}} = A_{\text{out}}H, \qquad \mathbb{E}[\widehat{\boldsymbol{x}}_{\text{out}} \mid \Theta] = R_{\text{out}}\boldsymbol{s} + A_{\text{out}}\boldsymbol{\mu}_0, \qquad C_{x,\text{out}} = A_{\text{out}} N A_{\text{out}}^{\mathsf{T}}. \tag{9}$$

The diagonal $D_{Q,\text{out}}$ contains only finite positive normalizations on $\mathcal{S}_{\text{out}}$, so its ordinary inverse is defined. The matrix A_{out} has no rows outside that set. Extending it to the full storage grid by writing zeros would be a representation convenience, not this scientific operator, response, or covariance.

Here N is the covariance of processed-sample error, not the ensemble covariance of sky plus error. The state Θ contains all selected or estimated calibration, alignment, membership, coefficients, response, geometry, identity, and effective-support facts. Equation (9) therefore makes a conditional statement on exactly the authorized rows. If any element of Θ is estimated from the same data, an unconditional claim requires its joint uncertainty and cross-covariances.

The response matrix $R_{\text{out}} = A_{\text{out}}H$ is the authority for what sky structure survives on the authorized output domain. A stored kernel image is the realized response to a named unit-bearing template $\boldsymbol{\tau}$ only if its sample-domain parent is $\boldsymbol{k} = H\boldsymbol{\tau}$ and the stored map is $A_{\text{out}}\boldsymbol{k} = R_{\text{out}}\boldsymbol{\tau}$ with the same exact row identity as signal. Without that identity it is a tracer, not the measured response. If the parent is missing, response is explicitly unavailable; no zero array can stand in for it.

V0.1 fixes the signal-centering operator to $L = I$. SCI-MAP therefore does not remove a mean or a constant mode during coaddition. It preserves a constant admitted sample vector on the exact support-authorized output rows because numerator and denominator use identical coefficients. A stronger constant-sky statement additionally requires the forward operator to map that sky mode to the admitted sample constant. Neither result alone establishes point-source, integrated-flux, extended-source, edge, or absolute astrometric fidelity. Any absolute mode already removed upstream remains absent and must be carried in response and null-mode provenance.

3 Input classes and falsifiable response behavior

The same response equation yields distinct, testable input classes without granting them equal scientific meaning:

Constant.
: Fixed positive coefficients preserve a constant admitted sample vector even when coverage and variance vary.

Delta or basis pixel.
: The output is a support-row-restricted column of R_{out}, including beam, projection, preprocessing, and boundary effects.

Point-source template.
: The output is $R_{\mathrm{out}}\boldsymbol{\tau}_b$ on exactly the authorized rows. Peak, integrated, and fit amplitudes are different functionals.

Resolved, gradient, and Fourier structure.
: The output is $R_{\mathrm{out}}\boldsymbol{\tau}$; constant preservation says nothing about these modes.

Variable coverage.
: Fixed-coefficient normalization may preserve a constant while covariance and nonconstant response vary spatially.

Finite boundary.
: Truncation can preserve a local constant yet change a compact-source peak or integral. No wrap or clamp is authorized.

One-hot and fractional projection must remain distinguishable. With one-hot assignment and calibrated independent inverse-variance coefficients, ordinary normalized gridding coincides with the corresponding diagonal weighted mean. For fractional projection, the ordinary denominator contains G_{pi}, while the diagonal least-squares normal term contains G_{pi}^2. They are not the same estimator. This difference also creates cross-pixel covariance when one sample contributes to multiple pixels.

Response fidelity is therefore a validation layer, not a consequence of code executing the algebra. Delta, point-template, extended, Fourier, variable- coverage, and edge injections are preregistered in Appendix D; they are not run or claimed here.

4 Conditional uncertainty and statistical labels

For independent selected sample errors with true inverse-variance coefficients, the fractional-projection specialization is:

$$\mathrm{Var}(\widehat{x}_p \mid \Theta) = \frac{\sum_i G_{pi}^2 \omega_i}{(\sum_i G_{pi}\omega_i)^2}, \qquad \mathrm{Cov}(\widehat{x}_p, \widehat{x}_r \mid \Theta) = \frac{\sum_i G_{pi} G_{ri}\omega_i}{Q_p Q_r}, \qquad p, r \in \mathcal{S}_{\mathrm{out}}, \tag{10}$$

$$\phi_p = \frac{Q_p^2}{\sum_i G_{pi}^2 \omega_i}, \qquad \phi_p = Q_p \iff \sum_i \omega_i (G_{pi}^2 - G_{pi}) = 0, \qquad p \in \mathcal{S}_{\mathrm{out}}, \tag{11}$$

Equation (11) states exactly why normalization is usually not precision. For $0 \le G_{pi} \le 1$ and strictly positive coefficients, the equality is guaranteed by one-hot contributions. A single pixel receiving two coefficients $G = 1/2$, $\omega = 1$ has $Q = 1$, variance $1/2$, and formal marginal inverse variance $\phi = 2$. The off-diagonal term in Equation (10) remains whenever pixels share sample support.

Four uncertainty objects must not be conflated:

1. Q, an ordinary normalization with its declared gridding unit;
2. $\phi = 1/[C_{x,\mathrm{out}}]_{pp}$, a conditional marginal inverse variance under its named assumptions;
3. $C_{x,\mathrm{out}}$ or $C_{x,\mathrm{out}}^{+}$, the full conditional covariance or precision on the identifiable stochastic subspace of the exact authorized row domain; and
4. an empirical covariance, weight, or significance calibrated by NOI.

The bundle must expose whether full covariance is persisted, lineage-resolvable, summarized, or unavailable, and must name omitted temporal, detector, cross-pixel, and calibration/response correlations. The minimum persistence form remains an owner decision. Calibration, astrometric, response, or additive bias nuisance terms are known, estimated with propagated uncertainty, or unknown; unknown is not zero. Signal times the square root of merely formal weight is formal standardized signal, never empirical `sig2noise`.

5 Eight facts, support, and raw validity

The raw bundle separates eight facts because they answer different questions:

Fact	Meaning
Geometric projected incidences	How many upstream-eligible candidate occurrences geometrically touched the pixel under the declared fractional counting convention.
Estimator-admitted contributions	How many occurrences passed every ordinary coefficient and contribution predicate.
Admitted observation count	How many complete observation bundles contribute to a coadd pixel; not sample hits or attempted observations.
Upstream-eligible exposure	Named exposure over the upstream-eligible projected population, before estimator-specific rejection.
Retained exposure	The same named exposure over estimator-admitted membership and placement.
Normalization support	The numerical threshold predicate used to authorize ordinary division.
Science-policy support	The separately stricter adopted policy predicate.
Final raw validity	The authoritative conjunction of admitted identity, support, finite signal and required companions, and no product failure.

No one fact is an alias for another. In particular, exposure is not precision, positive normalization is not final validity, and a finite payload is not evidence of scientific availability.

The adopted v0.1 threshold rule is policy authority, not a derived physical law. For the declared plane and policy population of finite positive Q_p values, sorted in ascending order, the admitted formula names $c = \texttt{coverage\_cut}$:

$$\mathcal{I}_{\Pi_{\text{eff}}} = \text{the declared support-policy population,} \tag{12}$$

$$\mathcal{P} = \{Q_p : \text{finite}(Q_p),\ Q_p > 0,\ p \in \mathcal{I}_{\Pi_{\text{eff}}}\}, \qquad N = |\mathcal{P}|, \tag{13}$$

$$k = \left\lfloor \frac{\lfloor 0.75N \rfloor + N}{2} \right\rfloor, \qquad Q_\star = \begin{cases} \mathcal{P}_{(k)}, & N > 0, \\ 0, & N = 0, \end{cases} \tag{14}$$

$$c = \texttt{coverage\_cut}, \qquad \text{admit}_{\Pi_{\text{eff}}}\big(c, \text{unit\_status}(c)\big) = 1, \tag{15}$$

$$T_{\text{norm}} = Q_\star c/10, \qquad T_{\text{sci}} = Q_\star c, \tag{16}$$

$$S_p^{\text{norm}} = \mathbf{1}[\text{finite}(Q_p),\ Q_p > 0,\ Q_p \geq T_{\text{norm}}], \tag{17}$$

$$S_p^{\text{sci}} = \mathbf{1}[\text{finite}(Q_p),\ Q_p > 0,\ Q_p \geq T_{\text{sci}}]. \tag{18}$$

Both predicates require the pixel's own explicit finite positive coefficient; the empty-population threshold of zero does not manufacture support. The operator row set $\mathcal{S}_{\text{out}}$ is the exact intersection of the predicate row sets designated applicable by Π_{eff}; it is not a full-grid mask with zero semantics.

The admitted packet supplies no numeric domain, range, or unit status for c. Equation (15) is therefore a precondition, not a claimed range: the owner-authorized effective policy must explicitly admit the exact value and its unit or dimensionless status. Any other state fails closed before support rows or a required product are constructed. In particular, this author draft does not decide negative, zero, non-finite, or greater-than-one values. Their disposition and the failure scope are SCI-MAP-OD-007. The physical rationale, future change authority, and evidence for replacing the policy remain SCI-MAP-OD-001 and SCI-MAP-OD-002. This contract records the admitted algorithm without inventing those answers.

6 Observation coaddition on one common grid

Coaddition consumes immutable complete observation bundles, not independent planes. Admission is atomic: unit, estimand, response state, frame/WCS, shape, group/Stokes identity, support/validity policy, covariance status, parentage, and required companions are checked before any coadd-owned state changes.

The only v0.1 placement is centered integer embedding on the same grid. Observation and coadd row and column differences are nonnegative and even; their half-differences are the integer offsets, and the shifted reference pixels identify the same world coordinate. The lossless typed WCS is the authority. No fractional shift, interpolation, reprojection, wrapping, or implicit source recentering is allowed.

For the atomically admitted per-pixel set $\mathcal{O}_p$, every contributing observation row is already a member of that observation's support-authorized scientific domain. Let $\mathcal{S}^c_{\mathrm{out}}$ be the exact rows authorized by the effective coadd support policy. For finite positive observation coefficients u_{op}, the reused ordinary coadd is:

$$\mathcal{S}^c_{\mathrm{out}} = \{p : \mathrm{authorize}_{\Pi^c_{\mathrm{eff}}}(p) = 1\}, \tag{19}$$

$$N^c_p = \sum_{o\in\mathcal{O}_p} u_{op}\widehat{x}_{op}, \qquad Q^c_p = \sum_{o\in\mathcal{O}_p} u_{op}, \tag{20}$$

$$\widehat{s}_p = N^c_p/Q^c_p, \qquad p \in \mathcal{S}^c_{\mathrm{out}}, \tag{21}$$

$$J^c_{\mathrm{out}}\boldsymbol{z} = (z_p)_{p\in\mathcal{S}^c_{\mathrm{out}}}, \tag{22}$$

$$\widehat{\boldsymbol{s}}_{\mathrm{out}} = (\widehat{s}_p)_{p\in\mathcal{S}^c_{\mathrm{out}}} = B_{\mathrm{out}}\widehat{\boldsymbol{x}}_{\mathrm{obs}}, \tag{23}$$

$$\widehat{k}^c_p = \frac{\sum_{o\in\mathcal{O}_p} u_{op}\widehat{k}_{op}}{Q^c_p}, \qquad p \in \mathcal{S}^c_{\mathrm{out}}. \tag{24}$$

$$C_{c,\mathrm{out}} = B_{\mathrm{out}}C_{\mathrm{obs}}B^{\mathsf{T}}_{\mathrm{out}}, \qquad R_{c,\mathrm{out}} = B_{\mathrm{out}}T. \tag{25}$$

The coadd operator B_{out} has rows only in $\mathcal{S}^c_{\mathrm{out}}$. Coadd accumulators or dense storage slots outside that set do not become a zero-valued scientific coadd. On its exact domain the coadd is a convex weighted mean. One admitted observation reproduces itself; unequal coefficients produce the corresponding weighted mean. Its normalization is not automatically precision. It is summed marginal inverse variance only when observations are unbiased unit-gain estimates of the same responded quantity, their errors are independent, and each u_{op} is the true marginal inverse variance. With cross-observation covariance, the exact conditional covariance retains those blocks on the same selected rows. A correlated-GLS benchmark may require negative combination coefficients, placing it outside this ordinary positive-coefficient contract.

Available response, retained exposure, and admitted observation count follow the same bundle admission and placement. A rejected observation changes no coadd state anywhere, even if a subset of its planes would otherwise appear compatible.

7 Identity, WCS, units, and lifecycle

A map cannot be reconstructed from its collection position. The minimum identity includes observation/coadd role; external observation or coadd ID; array and map group; network or occurrence-scoped detector binding where applicable; Stokes I; estimator; response state/version; signal unit; frame; full WCS; pixel shape/order and boundary policy; raw parentage; and exact product identity. Array ID, name, index, network, map group, map index, detector occurrence, and container position remain distinct.

Science maps use their declared equatorial J2000 TAN WCS with degree-valued spatial axes. A Pointing/OOF ordinary-map use, if the owner registers it, uses a declared AltAz tangent plane in arcseconds; its mode-specific meaning is outside SCI-MAP. FITS/WCS coordinates are one-based and in-memory row/column indices are zero-based.

Full-precision typed or sidecar WCS is the lossless admission and provenance authority. A physical FITS serialization may differ by at most 0.1 arcsec in maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. This is a representation tolerance, not authority for a different grid.

Requested state records the immutable accepted intent. Effective state records the executable supported plan. Observation-resolved state replaces all observation-specific arrays, mappings, geometry, coefficients, and provenance. Realized state records what actually ran, all accumulators and eight facts, products, offsets, representation states, cardinality, and failures. State flows one way; execution does not rewrite the request or inherit a prior observation's resolution.

8 Failure, downstream boundary, and immutable parentage

Failure semantics protect the scientific bundle from partial truth. An invalid candidate can be excluded at contribution scope; an unsupported pixel is raw-invalid; an incompatible observation is rejected atomically; an unrepresentable finite aggregate, coordinate/index conversion, missing required companion, or failed required publication propagates at its declared product scope before live state mutation. A completion marker cannot override a missing required output.

No downstream package receives a bare signal plane as a complete map. The boundary includes the estimator and normalization identity; response state; conditional covariance/formal-weight state and assumptions; eight facts; additive-bias and null-mode state; units; WCS; grouping; lifecycle; raw parentage; dependency versions; and failures. NOI may propagate a fixed operator but owns realization construction and empirical calibration. FLT owns filtered support, response, covariance, and validity. SRC, MODE, and BEAM own inference and fit uncertainty. FRUIT owns recurrence and iteration state.

Every consumer fails closed for information required by its claim. A response- dependent consumer rejects an unavailable response. A significance consumer rejects merely formal standardization. An absolute-level consumer rejects an unobservable offset. A downstream product may have local validity, but it cannot rewrite raw validity or sever the immutable raw-parent identity.

When an admitted noise realization is described as using the same operator, membership, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and coadd admission are fixed or identically prescribed. Recomputing them from randomized data creates a different estimator, regardless of matching array shapes or filenames.

9 Uncertainty, validity, validation, and claim layers

The contract deliberately separates four reasoning layers:

1. **Uncertainty layer:** conditional covariance and response are propagated through the fixed operators; omitted state uncertainty remains explicit.
2. **Validity layer:** contribution eligibility, numerical support, science-policy support, and final raw validity determine the exact rows on which the estimator is scientifically available; unsupported rows are absent, not zero-filled observations.
3. **Validation layer:** preregistered analytic fixtures, matrix comparisons, injections, covariance controls, representation round trips, and observational references test distinct claims.
4. **Readiness layer:** implementation evidence, operational reliability, data completeness, and release authority are assessed elsewhere.

Claim layer	Status here	Evidence needed before an achieved claim
Algebraic scientific contract	Proposed author draft	Scientific-owner review and contract freeze; analytic predictions are specified, not executed.
Engineering conformance	Not assessed	Requirement-by-requirement inspection and tests against one exact candidate identity.
Representation/response fidelity	Not assessed	WCS round trips and delta, template, extended, variable-coverage, and edge injections with preregistered bounds.
Observational performance	Not assessed	Independent astronomical amplitude, shape, position, covariance, and repeatability references where available.
Production readiness	Not assessed	Complete required artifacts, failure/log audit, operational gates, owner release decision, and no disqualifying gaps.

Floating reductions require a preregistered bound derived from operation count and conditioning. For a sum of n terms, a reference policy may use $\gamma_n = n\epsilon/(1 - n\epsilon)$ and a fixed multiple such as the admitted core's $8\gamma_n \sum_i |a_i|$, frozen before observing results. Integer counts, identities, masks, support, offsets, and product cardinality remain exact. Tolerances are not relaxed after failure.

The contract's 25 predictions are listed formally in Appendix D. No validation run, implementation inspection, observational reduction, or production decision was performed during this scientific authorship stage.

10 Open owner decisions and scientific disposition

Seven decisions remain intentionally unresolved: (1) the physical rationale for the adopted support thresholds; (2) future threshold change authority and evidence; (3) whether a typed unavailable response permits a scientifically usable bundle for restricted consumers; (4) the minimum covariance representation that must persist rather than remain lineage-resolvable; (5) whether physical observation-map publication is an abstract obligation beyond the effective v0.1 required-product rule; (6) whether Pointing/OOF ordinary arithmetic is a registered reuse of this operator; and (7) the admissible numeric domain, unit status, boundary-case dispositions, and failure behavior for $c =$ `coverage_cut`. Appendix A reproduces the exact questions and conservative interim behavior from the owner ledger.

Pending those decisions, the scientific contract is coherent at its stated conditional boundary. It authorizes ordinary positive-coefficient normalized mapmaking and atomic centered-integer observation coaddition only. It does not establish implementation conformity, representation fidelity, observational performance, or production readiness.

A Owner-decision register

This compact register is generated from `SCIENTIFIC_OWNER_DECISION_LEDGER.md`. The mechanical contract checker requires exact decision IDs, open statuses, decision prompts, and conservative interim dispositions to match before either PDF is accepted. Rendering this register does not answer or close an item.

Decision/status	Exact decision requested	Conservative draft behavior pending decision
`SCI-MAP-OD-001` **OPEN**	What physical rationale is authoritative for the adopted v0.1 normalization-support and science-policy-support threshold rule?	State the formula exactly, label it adopted policy, and make no physical optimality, completeness, noise, or response claim from it.
`SCI-MAP-OD-002` **OPEN**	Who may change the support-threshold rule after v0.1, and what evidence is required for a successor?	Freeze the v0.1 formula and population identity; treat any altered multiplier, quantile/index, population, or predicate as a contract change.
`SCI-MAP-OD-003` **OPEN**	Must every scientifically usable v0.1 map have a realized response/kernel product, or may a bundle with a typed unavailable response be used by a restricted class of consumers?	Require an explicit response state; never fabricate a response; all response-dependent consumers fail closed. Do not assert that a response-unavailable bundle is generally usable or unusable.
`SCI-MAP-OD-004` **OPEN**	What minimum covariance representation must persist in a raw bundle, and what may remain resolvable through lineage?	Require the exact conditional covariance equation, representation status, assumptions, omissions, and lineage. Stronger consumers fail closed when their needed covariance is unavailable.
`SCI-MAP-OD-005` **OPEN**	Beyond the adopted v0.1 effective required-output rule, is physical observation-map publication an abstract contract obligation whenever coaddition is requested?	Preserve the v0.1 rule that every product designated required by the effective plan, including required observation products during coaddition, must publish successfully. Do not generalize this into a future universal storage mandate.
`SCI-MAP-OD-006` **OPEN**	Is Pointing/OOF ordinary-map arithmetic a registered use of the shared SCI-MAP v0.1 operator, with mode-specific interpretation remaining outside this package?	State only that reuse is conditional on a fully conforming input bundle; grant no mode-specific fitting, astrometric, response, or production authority.
`SCI-MAP-OD-007` **OPEN**	What numeric domain and unit status are authoritative for `c=coverage_cut`, including the required disposition of negative, zero, non-finite, and greater-than-one values, and what failure behavior applies when the state is not authorized?	Assert no general domain or units. Require the exact value and unit or dimensionless status to be explicitly admitted by the owner-authorized effective support policy; otherwise fail closed before constructing support-authorized output rows or mutating a required product.

B Requirement and crosswalk summary

The detailed 52-row requirement crosswalk and 25-row prediction crosswalk remain in `CROSSWALK.md`. This compact index makes the rationale's traceability visible without duplicating the normative clause text or the detailed row table. Appendix C renders every canonical requirement, and Appendix D renders every prediction, directly from the shared authority.

Authority range	Scientific surface	Principal rationale / open gate
REQ-001	Version and revision identity	Cover; owner freeze
REQ-002–010	Capability, input, identity, projection, coefficients, lifecycle, and admission	Sections 1, 7, 8; upstream producer facts
REQ-011–018	Ordinary accumulators, exact support-row publication, response, and absolute mode	Sections 1–3; OD-003
REQ-019–024	Row-selected covariance, formal weight, nuisance state, and statistical labels	Sections 2, 4; OD-004
REQ-025–035	Eight facts, threshold policy, exact support rows, and raw validity	Section 5; OD-001, OD-002, OD-007
REQ-036–042	Complete bundles, atomic centered-integer coadd, row-selected coadd response/covariance, and GLS boundary	Sections 6, 8; OD-003, OD-004, OD-007
REQ-043–048	Product identity, WCS, provenance, failure, and publication	Sections 7–8; OD-005, OD-006
REQ-049–052	Immutable raw parent, consumer boundary, fixed realization operator, and claim separation	Sections 8–9; OD-003, OD-004, OD-006, OD-007
PRED-001–025	Analytic consequences and conformance/representation challenges	Sections 3, 9; Appendix D; not executed
OD-001–007	Exact unresolved scientific-owner questions	Section 10; Appendix A; all OPEN

C Canonical requirements

This appendix is rendered directly from the shared canonical LaTeX authority. Requirement identifiers are stable within SCI-MAP v0.1. Editing prose in this document does not silently change them.

SCI-MAP-REQ-001 – **Versioned authority.** The scientific contract version shall be v0.1. The rendered document revision shall be recorded separately as r0.1; changing layout or exposition alone shall not be represented as a new scientific contract version.

SCI-MAP-REQ-002 – **Capability and calibrated input.** The ordinary input shall be calibrated SCI-CAL ordinary-`xs` Stokes-I samples with exact sample-by-detector membership and active signal unit `mJy/beam`. No other stream, Stokes component, or accepted unit token shall inherit this authority.

SCI-MAP-REQ-003 – **Explicit identity.** Every candidate contribution shall carry explicit observation, array, network when applicable, map-group, Stokes, input-column, and occurrence/product-scoped detector-acquisition identity. Container position, row equality, or a local key alone shall not establish scientific identity.

SCI-MAP-REQ-004 – **Upstream eligibility.** SCI-MAP shall consume, without redefining, the upstream sample/detector eligibility, flag precedence, calibration-validity, and non-finite states supplied by VAL and SCI-CAL.

SCI-MAP-REQ-005 – **Projection authority.** Each contribution shall use an externally supplied finite sample-to-map relation in a declared frame, WCS, pixel basis, extent, and boundary convention. SCI-MAP shall not choose pointing, interpolate a missing coordinate, wrap, clamp, or invent an out-of-grid placement.

SCI-MAP-REQ-006 – **Ordinary coefficient status.** Each admitted ordinary analysis coefficient shall be finite and strictly positive, with declared unit, normalization scope, lifecycle, applied factors, and statistical status. Zero is noncontributing; negative and non-finite ordinary coefficients are invalid.

SCI-MAP-REQ-007 – **Exposure meaning.** Every exposure contribution shall be finite and nonnegative in a named unit and accounting convention. Detector-seconds shall not be labeled wall-clock integration time without an explicit multiplicity normalization.

SCI-MAP-REQ-008 – **Response input state.** The response input shall be either a sample-domain kernel/template tracer with source, normalization, unit, processing-parent identity, and membership, or an explicit typed unavailable state. A numerical sentinel shall not represent unavailability.

SCI-MAP-REQ-009 – **Immutable plan and lifecycle.** The accepted request shall be immutable and shall resolve one way through requested, effective, observation-resolved, and realized states. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

SCI-MAP-REQ-010 – **Contribution predicate.** For each pixel, the estimator contribution set shall be fixed by declared eligibility, finite signal, finite positive effective coefficient, admitted projection and boundary, and every required companion predicate before any payload is accumulated.

SCI-MAP-REQ-011 – **Ordinary accumulators.** On the fixed contribution set, SCI-MAP shall accumulate the pre-normalized signal numerator $N_p = \sum_i a_{pi} x_i$, normalization $Q_p = \sum_i a_{pi}$, and, when available, response numerator $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi} \omega_i > 0$.

SCI-MAP-REQ-012 – **Normalized publication.** The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{\text{out}}$ authorized by the applicable effective support policy. On those rows a finite $Q_p > 0$ gives $\widehat{x}_p = N_p / Q_p$ and an available response companion gives $\widehat{k}_p = K_p / Q_p$. Accumulators or storage slots may exist elsewhere, but no unsupported row, zero-filled value, or sentinel is a normalized scientific output or measured zero sky.

SCI-MAP-REQ-013 – **Accumulator interpretation.** The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

SCI-MAP-REQ-014 – **Constant preservation.** With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

SCI-MAP-REQ-015 – Projection-specific estimand. One-hot assignment and fractional projection shall be identified explicitly. For fractional projection, normalized gridding shall not be mislabeled diagonal weighted least squares, and Q_p shall not be promoted to formal precision by analogy.

SCI-MAP-REQ-016 – Same operator for linear companions. Signal, available response/kernel tracer, and declared linear realizations shall use the same admitted membership, geometric placement, and exact support-authorized output-row selection; retained exposure and coadd observation count shall use the same membership and placement wherever their definitions require them.

SCI-MAP-REQ-017 – Realized response claim. A stored kernel may be called the realized response to template $\boldsymbol{\tau}$ only when its sample parent satisfies $\boldsymbol{k} = H\boldsymbol{\tau}$ and the support-restricted map product satisfies $\widehat{\boldsymbol{k}}_{\text{out}} = A_{\text{out}}H\boldsymbol{\tau} = R_{\text{out}}\boldsymbol{\tau}$ on exactly the authorized rows. Otherwise it shall be labeled only a tracer or unavailable.

SCI-MAP-REQ-018 – Absolute-offset mode. SCI-MAP v0.1 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

SCI-MAP-REQ-019 – Conditional covariance. For fixed complete state Θ, the authoritative ordinary noise covariance on the exact support-authorized row set shall be $C_{x,\text{out}} = A_{\text{out}}NA_{\text{out}}^{\mathsf{T}}$, retaining consequential cross-pixel terms among those rows. It is conditional on selected calibration, pointing, eligibility, coefficients, response, geometry, and support policy; no zero-filled extension is scientific covariance.

SCI-MAP-REQ-020 – When normalization is precision. The normalization may be called marginal inverse variance only when coefficients are calibrated true inverse variances, selected sample errors are independent for the asserted domain, and the projection satisfies $\sum_i \omega_i(G_{pi}^2 - G_{pi}) = 0$; one-hot assignment is the standard sufficient case.

SCI-MAP-REQ-021 – Formal diagonal weight. When its assumptions hold on an authorized output row, the formal marginal inverse variance shall be $\phi_p = Q_p^2/\sum_i G_{pi}^2\omega_i$. It shall remain distinct from Q_p, from the support-restricted full precision matrix $C_{x,\text{out}}^{+}$, and from empirical statistical weight.

SCI-MAP-REQ-022 – Covariance and nuisance state. The bundle shall declare whether full covariance is persisted, lineage-resolvable, summarized, or unavailable; it shall list omitted correlations and the status of calibration, astrometric, response, and additive-bias nuisance terms. Unavailable information shall not be silently set to zero.

SCI-MAP-REQ-023 – Data-derived state. If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

SCI-MAP-REQ-024 – No significance promotion. Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not `sig2noise` or empirical significance. Empirical calibration remains NOI-owned.

SCI-MAP-REQ-025 – Eight distinct facts. Every raw bundle shall separately represent geometric projected incidences, estimator-admitted contributions, admitted observation count for coadds, upstream-eligible exposure, retained exposure, numerical normalization support, science-policy support, and authoritative final raw science validity. No alias shall become authority for another fact.

SCI-MAP-REQ-026 – Geometric incidences. Geometric projected incidences shall count upstream-eligible entries whose declared projection touches a pixel, using an explicit fractional-projection counting convention, regardless of whether a positive ordinary coefficient is later admitted.

SCI-MAP-REQ-027 – Estimator contributions. Estimator-admitted contributions shall count exactly the entries in the ordinary contribution set and shall exclude zero, negative, non-finite, flagged, and out-of-bounds candidates.

SCI-MAP-REQ-028 – Admitted observation count. For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

SCI-MAP-REQ-029 – Upstream-eligible exposure. Upstream-eligible exposure shall sum the named exposure measure over the externally eligible projected population, before estimator-specific coefficient rejection, according to its declared exposure projection.

SCI-MAP-REQ-030 – Retained exposure. Retained exposure shall sum the same named exposure measure over estimator-admitted membership and placement; it shall not be labeled precision.

SCI-MAP-REQ-031 – Adopted threshold selector. For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p; for count $N > 0$, select zero-based $k = \lfloor(\lfloor 0.75N\rfloor + N)/2\rfloor$ and $Q_\star = Q_{(k)}$; for empty input set $Q_\star = 0$. The exact population identity, exact $c =$ `coverage_cut`, and its unit or dimensionless status shall be recorded. The effective support policy shall explicitly admit that exact c state before support resolution; otherwise the required product shall fail closed.

SCI-MAP-REQ-032 – Separate support predicates. Only after the effective support policy explicitly admits the exact value and unit status of c, numerical normalization support shall require an explicit finite positive $Q_p \geq Q_\star c/10$, and science-policy support shall independently require an explicit finite positive $Q_p \geq Q_\star c$. This contract asserts no general numeric domain or units for c. Every non-authorized state fails closed before a support row set is constructed. Neither support state shall establish final validity or precision.

SCI-MAP-REQ-033 – Authoritative raw validity. One raw science-validity state shall require membership in the exact support-authorized output-row set, admitted bundle identity, both applicable support predicates, finite normalized signal, finite required companions, and absence of a product-level failure. A retained slot, finite payload, positive coefficient, exposure, hit count, support plane, or compatibility alias alone shall not make a pixel valid.

SCI-MAP-REQ-034 – Screen before arithmetic. Invalid contributions shall be excluded before their numerical payload is evaluated. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

SCI-MAP-REQ-035 – Exceptional contribution states. Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

SCI-MAP-REQ-036 – Complete observation bundle. An observation map shall be one immutable ordered bundle containing signal, numerator, normalization identity, response state, conditional uncertainty state, all eight facts where applicable, units, frame/WCS, shape/indexing, estimator, lifecycle, parentage, and exact product identity.

SCI-MAP-REQ-037 – Atomic coadd admission. A complete observation bundle shall pass all compatibility and required-companion checks before any coadd-owned state changes. Rejection shall leave numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

SCI-MAP-REQ-038 – Centered integer common grid. Observation and coadd shapes shall differ by non-negative even row and column counts, with the corresponding integer offset mapping their reference pixels to the same world coordinate. Fractional shift, reprojection, interpolation, wrapping, implicit recentering, and a different sky grid are forbidden.

SCI-MAP-REQ-039 – Ordinary coadd estimator. For atomically admitted bundles and finite positive observation coefficients u_{op}, the coadd shall accumulate $N^c_p = \sum_o u_{op}\widehat{x}_{op}$ and $Q^c_p = \sum_o u_{op}$, then publish $\widehat{s}_p = N^c_p/Q^c_p$ only for rows in the exact effective coadd support set $\mathcal{S}^c_{\mathrm{out}}$. No unsupported coadd row or zero-filled substitute is a scientific output.

SCI-MAP-REQ-040 – Coadd companions. Available response/kernel shall use the same atomic admission, centered-integer placement, and exact support-authorized output-row selection as coadd signal. Retained exposure and admitted observation count shall use the same admission and placement; the count increments once per admitted complete bundle at each applicable pixel.

SCI-MAP-REQ-041 – Coadd covariance and precision. For fixed admission and support-restricted coadd operator B_{out}, conditional covariance shall be $C_{c,\mathrm{out}} = B_{\mathrm{out}} C_{\mathrm{obs}} B_{\mathrm{out}}^{\mathsf{T}}$, including consequential cross-observation blocks among authorized rows. Summed observation coefficients equal coadd precision only for independent unbiased maps of the same unit-gain estimand weighted by their true marginal inverse variances.

SCI-MAP-REQ-042 – Correlated-GLS boundary. A correlated minimum-variance solution using $K_p^{-1}\mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd.

SCI-MAP-REQ-043 – Minimum product identity. Every observation and coadd product shall identify role, external observation or coadd identity, array and group, network/detector occurrence when applicable, Stokes I, estimator, response state/version, signal unit, frame, WCS, pixel shape/order, boundary policy, parentage, and product identity.

SCI-MAP-REQ-044 – Frames and units. Science products shall use their declared equatorial J2000 TAN WCS with degree-valued spatial axes; Pointing/OOF registered uses, if approved, shall use a declared AltAz tangent plane in arcseconds. Signal and a unit-bearing realized response carry `mJy/beam`; covariance carries its square and a conditionally valid formal weight its inverse square.

SCI-MAP-REQ-045 – WCS and index persistence. Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. A physical FITS WCS may differ by at most 0.1 arcsec maximum sky separation while preserving exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations. FITS coordinates are one-based and memory indices zero-based.

SCI-MAP-REQ-046 – Realized provenance. Realized state shall record executed observations and groups, admitted contributions and bundles, accumulators, exact observation and coadd support-authorized output-row sets, all eight facts, response and covariance representation states, centered-integer offsets, completed required products, failures, and the exact requested/effective/resolved parents.

SCI-MAP-REQ-047 – Failure scope and atomicity. An invalid required input, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing required companion, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before live aggregate state changes.

SCI-MAP-REQ-048 – Required observation products. A complete observation bundle remains required when coaddition is enabled, and every product designated required by the effective v0.1 plan shall be published successfully before completion is asserted. Whether future abstract contracts require physical observation-map publication independent of an effective product policy remains an owner decision.

SCI-MAP-REQ-049 – Immutable raw parent. Downstream transformations shall preserve resolvable raw bundle identity and the authoritative raw validity state. They shall not rewrite raw validity or promote a raw-invalid pixel through a filtered or consumer-local validity alias.

SCI-MAP-REQ-050 – Complete consumer boundary. No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include estimator/normalization identity, response state, conditional uncertainty state and assumptions, eight facts, additive-bias/null-mode state, units, frame/WCS, shape/indexing, grouping, lifecycle, parentage, dependency versions, and failures; each consumer shall fail closed for missing facts needed by its claim.

SCI-MAP-REQ-051 – Same fixed noise operator. An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

SCI-MAP-REQ-052 – Method and claim separation. This authority shall not be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

D Falsifiable predictions

These predictions are preregistered contract consequences or conformance challenges. Their presence is not validation evidence.

SCI-MAP-PRED-001 – Uniform constant. For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized raw-valid output row; numerator and normalization scale together, and no unsupported scientific row is created. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-002 – Unequally weighted constant. For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-003 – One-pixel hand calculation. For an authorized output row with admitted values x_i and coefficients $a_i > 0$, the exact expected triplet is $(N, Q, \widehat{x}) = (\sum a_i x_i, \sum a_i, \sum a_i x_i / \sum a_i)$; for an unauthorized row only declared accumulators or storage state may remain, never a normalized scientific zero. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-004 – Fractional split. One sample split with $G_{p1} = G_{r1} = 1/2$ contributes to both pixels, creates cross-pixel covariance, and distinguishes normalized gridding from diagonal WLS. Two equal independent split samples give $Q = 1$, variance $1/2$, and $\phi = 2$ per affected pixel. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-005 – Delta input. A basis-pixel sky input produces the corresponding support-row-restricted column of $R_{\mathrm{out}} = A_{\mathrm{out}} H$, including projection and boundary truncation; it need not be a delta in the output map, and no value is predicted outside the authorized rows. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-006 – Point-source template. A declared point-source template $\boldsymbol{\tau}_b$ produces $R_{\mathrm{out}} \boldsymbol{\tau}_b$ on exactly the support-authorized rows; peak, integrated, and fitted amplitudes are distinct functionals and need separate criteria. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-007 – Resolved and Fourier inputs. Resolved disks, gradients, and Fourier modes produce their explicit $R_{\mathrm{out}} \boldsymbol{\tau}$ on the authorized row set; a constant-input pass does not predict preservation of these modes. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-008 – Variable coverage. Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-009 – Finite boundary. At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes. *Primary claim layer: Representation fidelity.*

SCI-MAP-PRED-010 – Coefficient states. Zero coefficients add no estimator contribution; low positive coefficients follow the adopted support policy; negative and non-finite ordinary coefficients are invalid and cause-specific. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-011 – Eligibility and non-finite states. Flagged, non-finite, and out-of-bounds candidates do not contaminate any accumulator because they are excluded before payload evaluation; zero or failed support produces no scientific output row rather than measured zero. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-012 – Threshold selector. For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor (\lfloor 0.75N \rfloor + N)/2 \rfloor$; empty input yields threshold zero, but no pixel gains support without its own finite positive coefficient. Any exact c value or unit state not explicitly admitted by the owner-authorized effective policy fails closed before a support row set is constructed; no numeric domain is inferred. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-013 – Missing response. A missing response parent yields a typed unavailable response state, never a fabricated zero response; any response-dependent consumer rejects it pending the owner decision on restricted usability. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-014 – Atomic rejection. A missing required companion or any incompatible identity, unit, response, WCS, shape, policy, or parent rejects the entire observation before every coadd accumulator and count, including apparently compatible pixels, is changed. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-015 – One-observation coadd. A coadd containing one admitted observation with positive coefficient reproduces that observation's signal and available response exactly on the corresponding centered-integer rows authorized by both the observation and coadd support policies; it creates no unsupported scientific row. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-016 – Unequal observation coefficients. On each support-authorized coadd row, two compatible observations produce the positive-coefficient weighted mean and the summed normalization; the result remains inside the convex hull of their finite values, while an unauthorized row has no normalized coadd value. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-017 – Missing or invalid observation. An unsupported observation row contributes at no coadd row; an incompletely admitted observation contributes nowhere and does not increment admitted-observation count. Neither case creates a zero-valued scientific substitute. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-018 – Centered integer placement. Compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-019 – Cross-observation covariance. For two maps with variances σ_1^2, σ_2^2 and covariance c, the minimum-variance coefficients depend on c; summing marginal normalizations is not precision unless the required independence and calibration conditions hold. Covariance comparison uses the exact support-authorized observation and coadd rows. *Primary claim layer: Algebraic contract.*

SCI-MAP-PRED-020 – Fixed noise operator. A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient A_{out} and B_{out} as the signal path; recomputed state exposes a named difference. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-021 – Sequential and parallel reduction. Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-022 – No false significance. A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-023 – No validity promotion. A downstream product may have its own local validity but cannot change an inherited raw-invalid pixel to raw-valid or sever its raw-parent identity. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-024 – No ordinary predicate for JINC. A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.1. *Primary claim layer: Engineering conformance.*

SCI-MAP-PRED-025 – Aggregate and index failure. An exact-zero, non-finite, overflowed, or unrepresentable aggregate/index never becomes a live valid product; failure occurs at the declared scope before mutation or completion. *Primary claim layer: Engineering conformance.*

E Product vocabulary

Product	Unit	Contract meaning
Signal numerator	coefficient times `mJy/beam`	Pre-normalized accumulator; not a sky estimator.
Normalization	declared gridding unit	Positive denominator; precision only under Requirements 020–021.
Normalized signal	`mJy/beam`	Conditional ordinary map estimand whose vector contains exactly the effective-policy-authorized output rows; no unsupported zero-filled semantic extension.
Formal diagonal weight	inverse square signal unit	Conditional marginal inverse variance under named assumptions; not empirical significance.
Conditional covariance	square signal unit	Full fixed-state noise propagation on the exact scientific output-row domain, with omissions and representation status named.
Response/kernel	`mJy/beam` when unit-bearing	Realized template response only when the sample-parent identity is proven; otherwise tracer or unavailable.
Geometric/estimator hits	integer	Counts of declared populations, not coverage, precision, or validity.
Exposure facts	named time/accounting unit	Upstream-eligible and retained accounting measures, not precision.
Support facts	logical	Separate normalization and science-policy predicates; their effective conjunction defines output-row membership.
Raw validity	logical	Authoritative consumer-eligibility state for the raw bundle, preserving required identity and companion finiteness.

F Admitted source register and independence statement

The scientific author used only the following admitted sources:

Source	Exact admitted identity
Scope Brief	`SCOPE_BRIEF.md` SHA-256 `e2a9eb51edb5956191813b4cdbd23866e875d52cdf89cd8b6c272988b4f26674`.
Supersession Cover	`AUTHOR_SUPERSESSION_COVER.md` SHA-256 `8ea283525f18199d9760c3f672d145d71f0db87b320ab2e88a5c6635ef3d4aa0`.
Frozen core	Exact Git object `c28f18ed089657dae278caba2d6d6d65c7ec72f4:doc/audits/packages/SCI-MAP-001_INDEPENDENT_CORE.tex` SHA-256 `13dd5922bd492e381afcc3b015284216dde1ccc2199ece3d070ee577c7324381`.
Conventions and ownership	`AUTHOR_CONVENTIONS_AND_OWNERSHIP.md` SHA-256 `2d478cb6c5e897308d19614b8b016633187449718 50c67459f84c7ddcd57c5c9`.

No implementation, audit, finding, repair, test, validation, reduction, production-status record, MAP-002/MAP-003 evidence, or web source entered this authorship. The frozen ordinary derivation was consolidated rather than rederived for novelty.